

Esperanza condicionada a σ -álgebras anidadas

Proposición 1. Sea $X \in \mathcal{L}^1(\mathbb{P})$ y $\mathcal{F}_1, \mathcal{F}_2$ σ -álgebras tales que $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \Sigma$

$$\implies \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_1] \mid \mathcal{F}_2] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_2] \mid \mathcal{F}_1] = \mathbb{E}[X \mid \mathcal{F}_1] \text{ c.s.}$$

Demostración: Como $\mathcal{F}_1 \subset \mathcal{F}_2$, entonces $\mathbb{E}[X \mid \mathcal{F}_1]$ es $\mathcal{F}_2$ -medible. Por lo tanto, por la propiedad `lem-esperanza-condicionada/(3)` del `lem-esperanza-condicionada/Lema 1`, tenemos que

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_1] \mid \mathcal{F}_2] = \mathbb{E}[X \mid \mathcal{F}_1] \text{ c.s.}$$

Para la otra igualdad, sea Z $\mathcal{F}_1$ -medible, entonces también es $\mathcal{F}_2$ -medible.

$$\implies \mathbb{E}[Z \cdot \mathbb{E}[X \mid \mathcal{F}_1]] \stackrel{Z \text{ es } \mathcal{F}_1\text{-medible}}{\stackrel{\downarrow}{=}} \mathbb{E}[X \cdot Z] \stackrel{Z \text{ es } \mathcal{F}_2\text{-medible}}{=} \mathbb{E}[Z \cdot \overbrace{\mathbb{E}[X \mid \mathcal{F}_2]}^{\mathcal{F}_2\text{-medible}}] \stackrel{Z \text{ es } \mathcal{F}_1\text{-medible}}{\stackrel{\downarrow}{=}} \mathbb{E}\left[Z \cdot \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_2] \mid \mathcal{F}_1]\right]$$

Por tanto, por unicidad de la esperanza condicionada, queda demostrado. ■

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